

Report Lab 12: Number extraction

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Gen AI tool used: GPAI <https://gpai.app>, ChatGPT

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Initial Prompt:

String parsing class in C++ for extracting embedded double-precision numbers from user string inputs on the command line. Functional requirements are:

1. Must extract any integer, double, scientific notation.
2. Must return and display a double.
3. Assume valid inputs will never produce the value -999999.99.
4. Scientific notation: the base can be either an integer or double, exponent must be an integer, exponent can be positive or negative.
5. Input is a single line of text up to 200 characters.
6. Cannot use library functions such as `std::stod`, `atof`, or `strtod`. May use standard string functions like `isdigit`, `isalpha`.
7. Output is formatted as "Extracted number: #####.#####" for valid input or "Invalid input: no valid floating-point number found" for invalid input. Can use `std::fixed` and `std::setprecision(4)`.
8. main functions continuously prompts user for input until user enters END (case-sensitive).
9. main function is the class driver and the function "double extractNumeric(const std::string& str)" is extraction logic.
 - 10)extractNumeric scan's left to right, ignores garbage characters, successful returns the number as a double, unsuccessful returns -999999.99
10. Valid range of numbers from approximately -1.7e308 to 1.7e308, if outside of range return -999999.99.
11. 11) Valid range of numbers from approximately -1.7e308 to 1.7e308, if outside of range return -999999.99.

Produced: NumericParser.cpp

Edited:

- Line 108: Added to display "Program terminated." to match the Sample Run.
- Line 118: Changed if condition from `(std::abs(result - (-999999.99)) < 1e-9)` to `(result == -999999.99)`.
 - Result is validated in `extractNumeric` so there's no need to validate it again.

- Line 89: If condition (std::isinf(value) || std::isnan(value)) changed to (std::isinf(value) || std::isnan(value) || value < -999999.99 || value > -999999.99)
 - Changed to match Sample Runs
- Line 123: Added to add blank lines in between prompts to match Sample Run.

Followup prompts:

1)Generate a makefile

Produced a Makefile.

Edited:

- Line 9: "SRCS = main.cpp" changed to "SRCS = NumericParser.cpp".
 - All code is contained in NumericParser.cpp
- Line 29: ".PHONY all clean"
 - Removed

2) 12.34.56 Is processed as a valid number when it should be invalid

- Line 40: "break;" changed to "return INVALID_VAL;"
 - Returns invalid as soon as it processes two decimals in a single number to match Sample Run. This would need to be updated if we need to be able to continue processing input to look for valid numbers beyond this fail condition.

3) In ChatGPT: 1e+ should be invalid:

- Line 82: "i = expStart;" changed to "return INVALID_VAL;"

Verification:

All code has been reviewed and tested. I removed unnecessary code as detailed above and couldn't find any other redundant code or other errors. Only concern is right now is input such as "12.34.56-784", the first invalid number hides the second valid number so whether this should be accounted for or not. Provided screenshot below shows test runs matching the sample runs.

[illegible]